Machine Learning

Logistic Regression

Dong-Kyu Chae

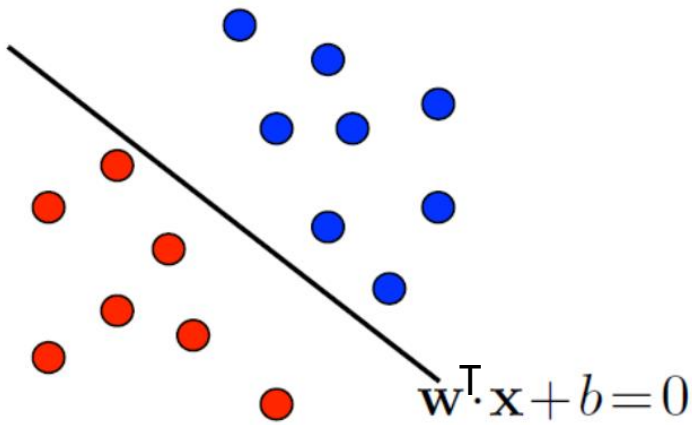
PI of the Data Intelligence Lab @HYU
Department of Computer Science
Hanyang University

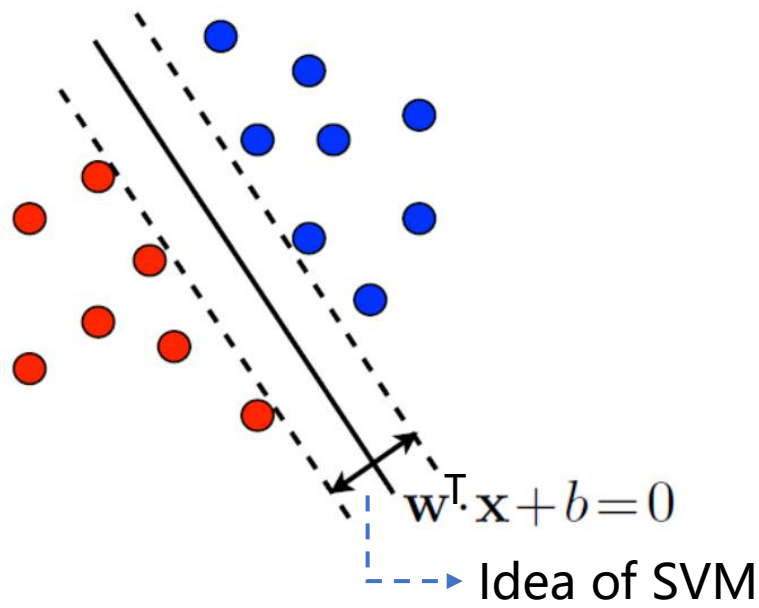
Data
Intelligence
LAB

Machine Learning with Linear Models

- **How to classify data with a linear function?**
  - Function: $g(\mathbf{x}) = \mathbf{w}^{\mathbf{T}}\mathbf{x} + b = 0$

    = Model, machine, decision boundary, etc…



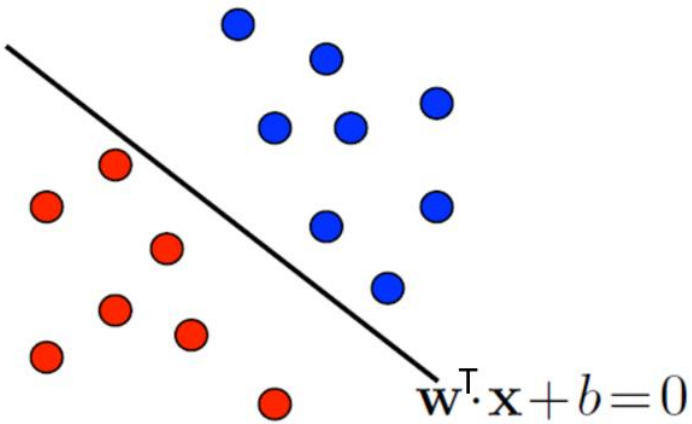


  - Classification of a data $\mathbf{x}_{\boldsymbol{n}}$: $sign(g(\mathbf{x}_{\boldsymbol{n}}))$

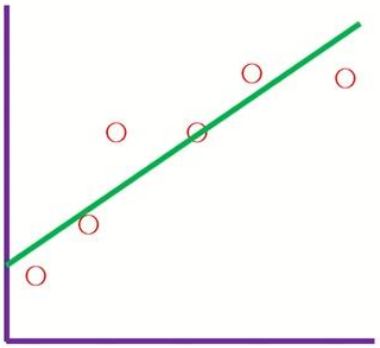
$$\mathbf{x} \rightarrow sign(\mathbf{w}^{\mathsf{T}}\cdot\mathbf{x} + b : \ \mathbf{w} \in R^d, \ b \in R)$$

Machine Learning with Linear Models

- **Linear classification**
  - Function: $g(\mathbf{x}) = \mathbf{w}^{\mathbf{T}}\mathbf{x} + b = 0$
  - Classification: $sign(\mathbf{w}^{\mathbf{T}}\mathbf{x}_n + b)$
    - If $sign(\quad) \geq 0$: class $y = \mathbf{1}$ (or, "blue")
    - If $sign(\quad) < 0$: class $y = \mathbf{0}$ (or, "red")



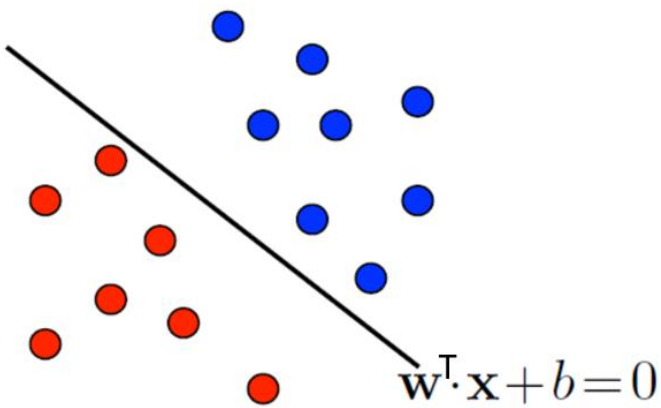
- **Linear regression**
  - $y = \mathbf{w}^{\mathbf{T}}\mathbf{x} + b$

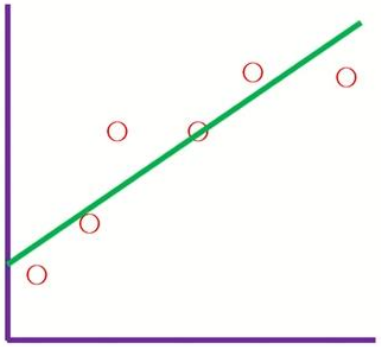


Machine Learning with Linear Models

Overview

- They deal with signal: $s = \mathbf{w}^{\mathbf{T}}\mathbf{x} + b$ in common
- Linear regression: signal itself == output, for predicting real values
  - $y = s = \mathbf{w}^{\mathbf{T}}\mathbf{x} + b$
- Linear classification: signal is thresholded at '0' to produce +1, -1 output
  - If $sign(s) \geq 0$: $y = \mathbf{1}$
  - If $sign(s) < 0$: $y = \mathbf{0}$





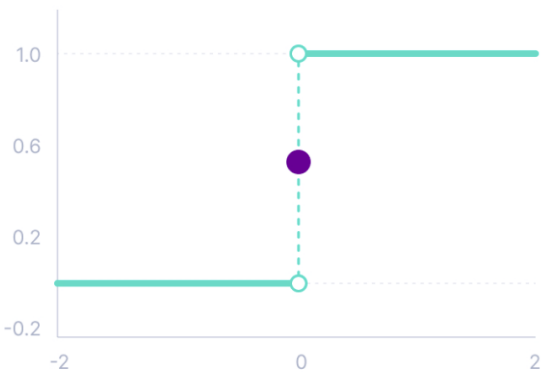
Motivation of Logistic Regression

- **Classification with probability**
  - In many situations, we cannot predict something with any certainty
    - Predicting cancer in the future
    - Stock / Bitcoin price prediction
    - Weather prediction
    - …
  - We would like to show "how likely something is to occur" given situations

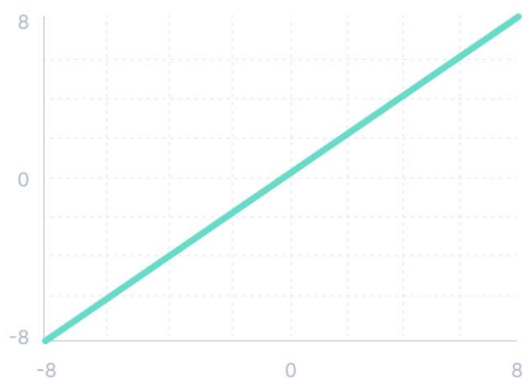
- **Need a more suitable model than the binary decision!**
  - An output y varies continuously between 0 and 1
  - The closer y is to 1, the more likely something will occur

Logistic Regression

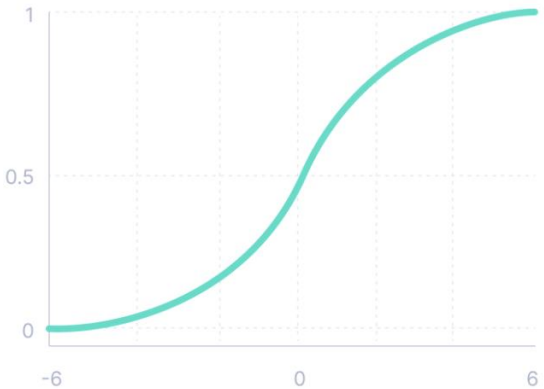
- **"Soft" binary classification**
  - It returns soft labels (probability, 0 ~ 1)
  - It is probability regression: it smoothly restricts output to probability range: [0, 1]



< **hard** binary classification>



<linear regression (unbounded)>

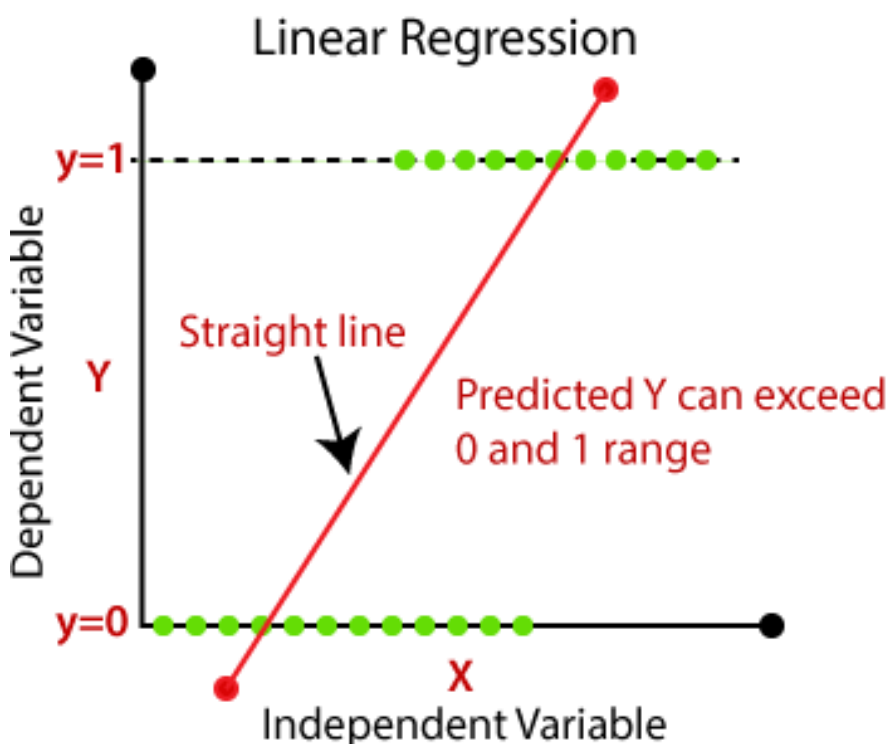


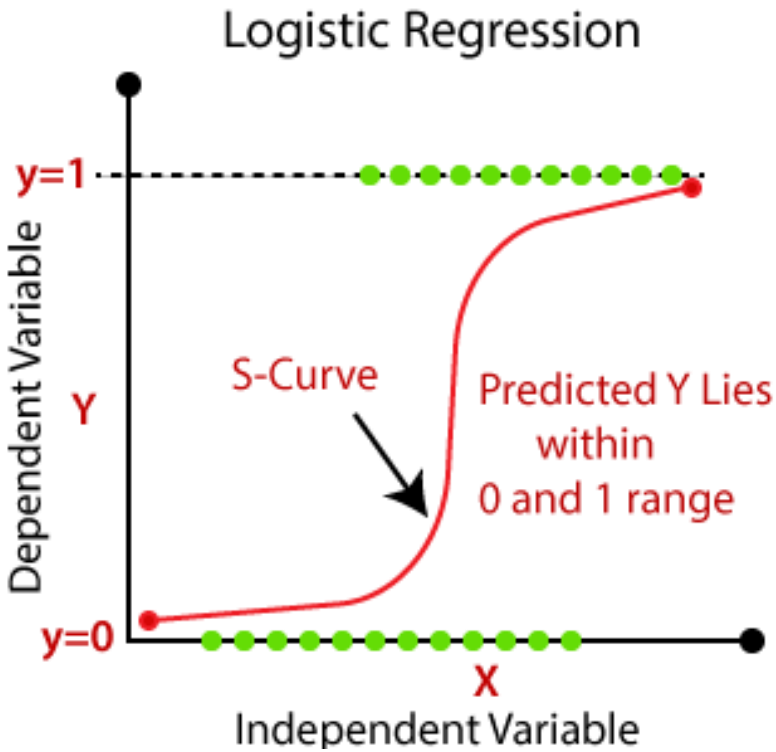
<logistic regression (bounded)>

- **Logistic regression**
  - Output: **real** (like regression), but **bounded** (like classification)
  - It allows uncertainty: intermediate values between 0 and 1 reflects the uncertainty

Logistic Regression

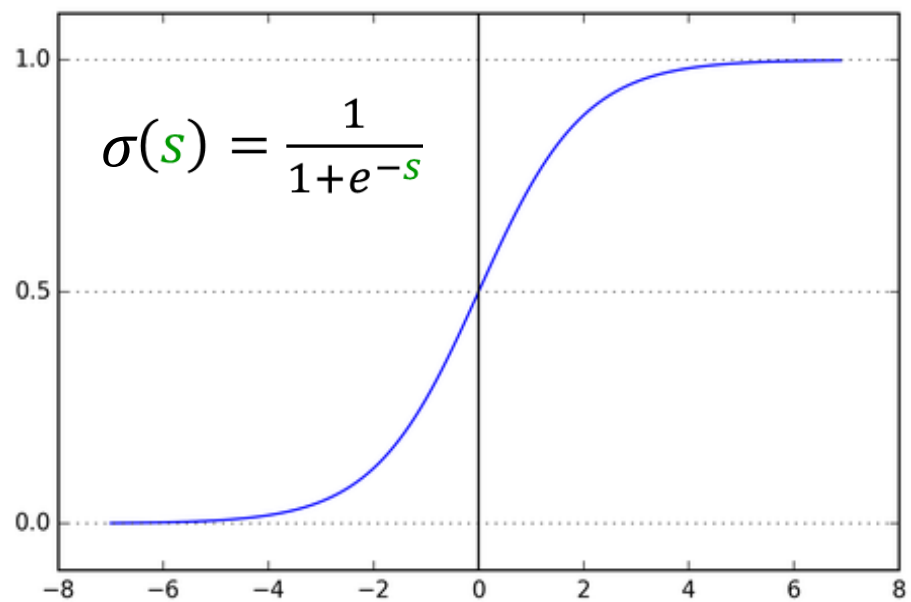
- **How to restrict regression model output to [0, 1]?**
  - Put our signal $s = \mathbf{w}^{\mathrm{T}}\mathbf{x} + b$ into **some function** bounded within 0 and 1
  - We call that kind of function as "sigmoid function"





Logistic Regression

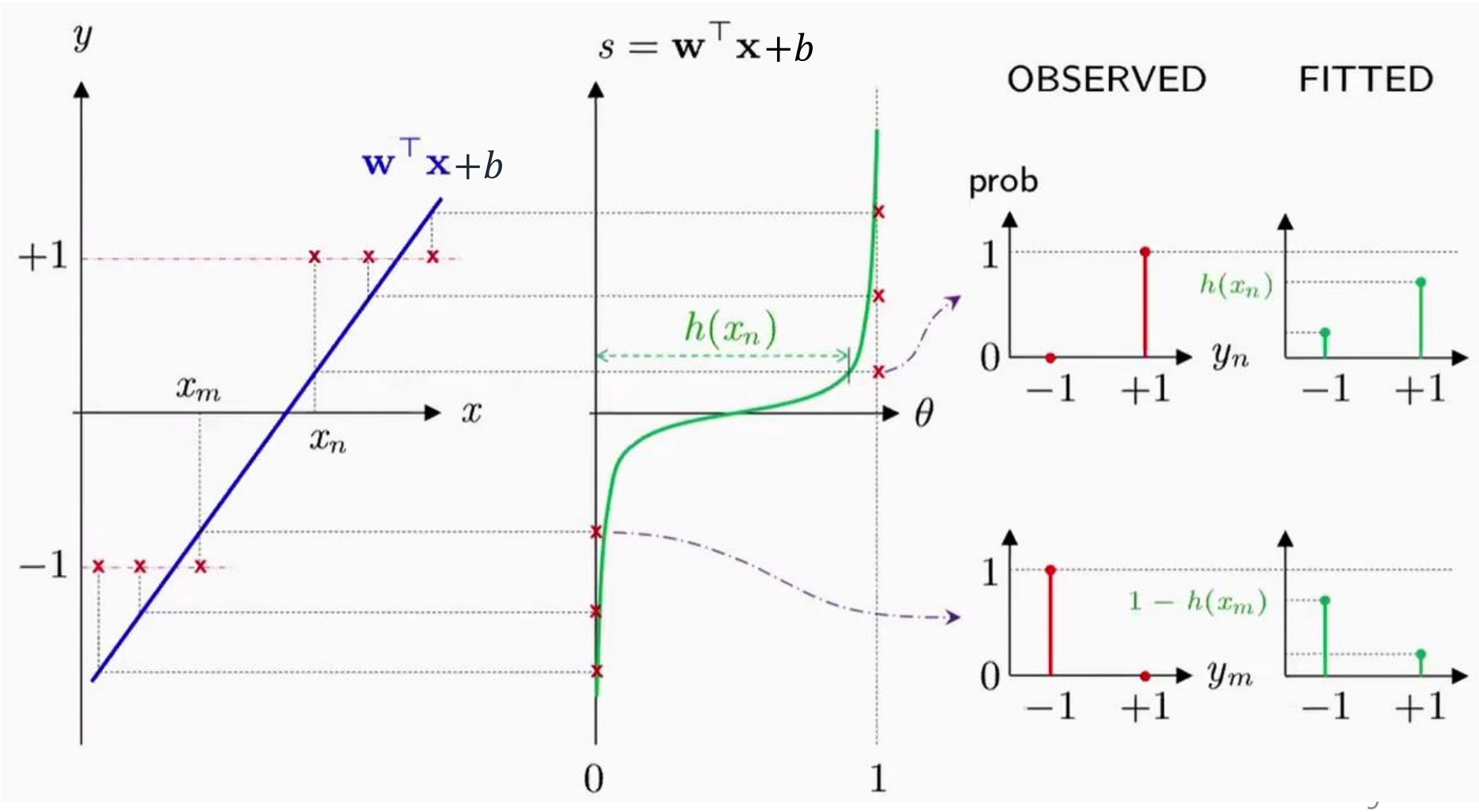
- **Logistic function $\sigma(s)$ (a.k.a. sigmoid)**
  - Input: s, $[-\infty, +\infty]$
  - Output: $\sigma(s) = \frac{1}{1+e^{-s}}$, [0, 1]
  - Output lies between 0 and 1 => can be interpreted as **probability** for binary events



- **Notes**
  - $\frac{1}{1+e^{-s}} = \frac{e^{s}}{1+e^{s}}$
  - $1 - \sigma(s) = 1 - \frac{1}{1+e^{-s}} = \frac{e^{-s}}{1+e^{-s}} = \sigma(-s)$

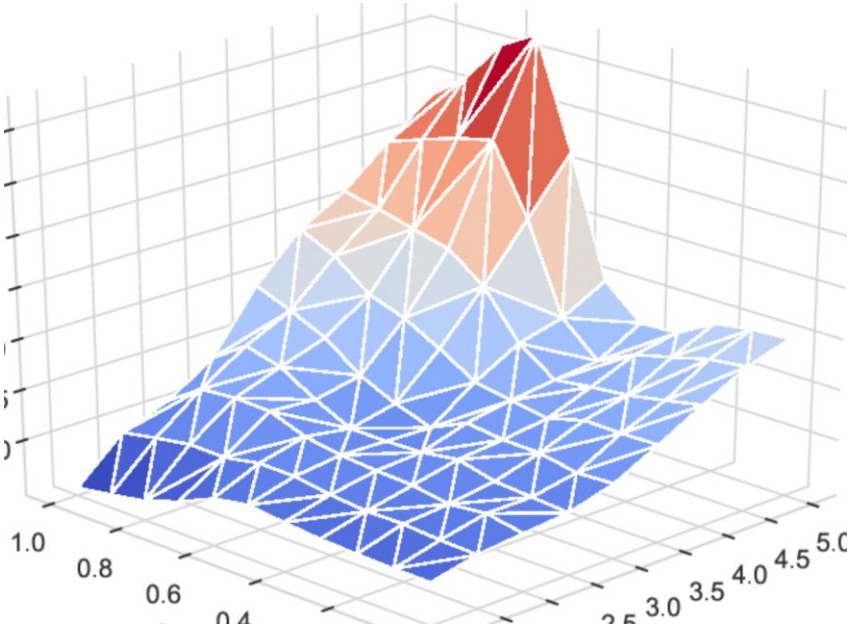
Logistic Regression

❑ Overview



Logistic Regression

- **Training logistic regression**
  - Model prediction: $\sigma(s)$
    - $\sigma(s) = \frac{1}{1+e^{-s}}$
    - $s = \mathbf{w}^{\mathbf{T}}\mathbf{x} + b$
  - Actual output: $y = \mathbf{1}$ or $\mathbf{0}$
  - Loss: **MSE…?**
    - $L = \frac{1}{N}\sum_{i=1}^{N}\left(y_i - \sigma(s_i)\right)^2$

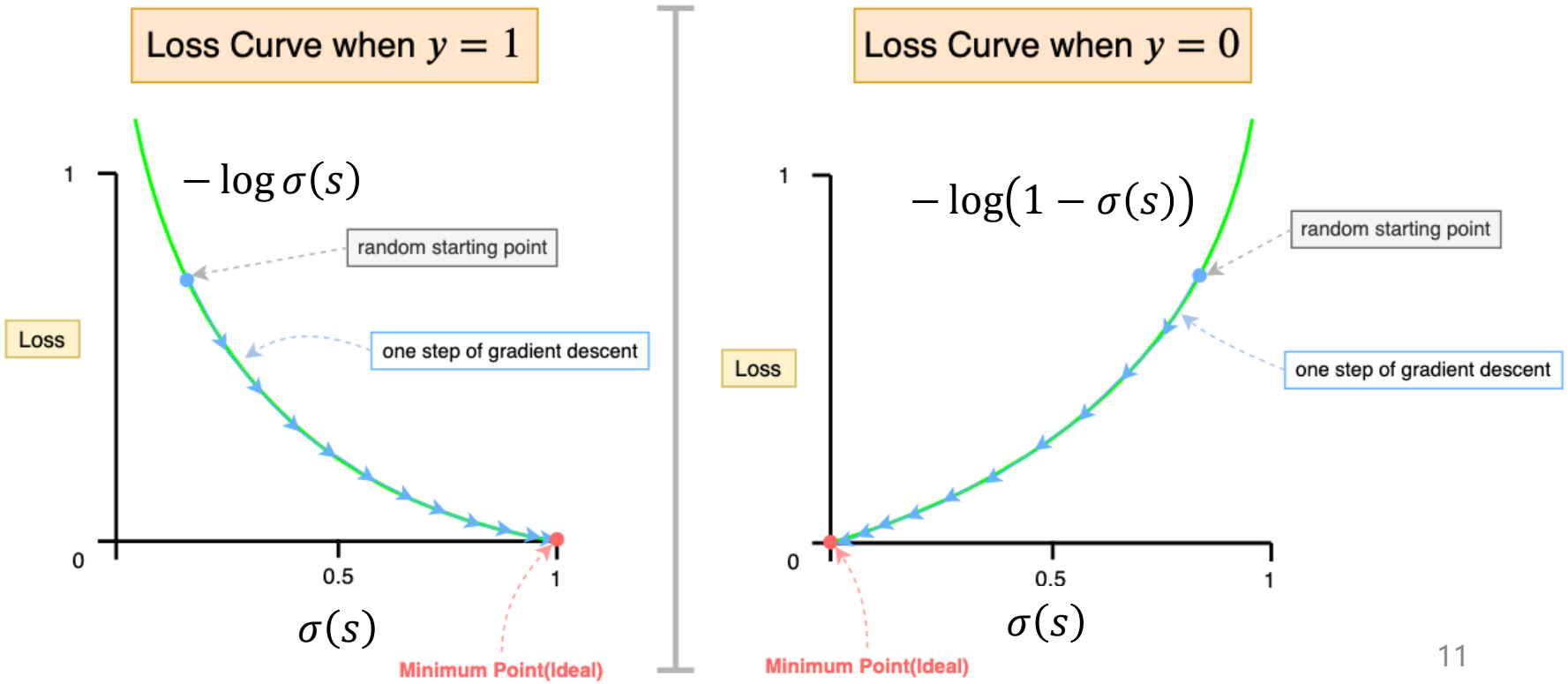


    - It is a square of non-linearity, which results in too complicated loss function surface
    - It may work, but is not effective

Logistic Regression

Objective function for logistic regression

- Log loss:
  - $\begin{cases} -\log\sigma(s) & \text{, if } y == 1 \\ -\log\big(1-\sigma(s)\big) & \text{, if } y == 0 \end{cases}$: two types of model predictions



Logistic Regression

- **Objective function for logistic regression**
  - Combining the two loss terms together

$$L = -y \cdot \log\sigma(s) - (1-y) \cdot \log\big(1-\sigma(s)\big)$$

  - This is called as **binary cross-entropy loss**
    - It computes difference between two given probability distributions

    : actual output [y, $1-y$] VS prediction [$\sigma(s)$, $1-\sigma(s)$]

      - E.g., [1,0] VS [0.72, 0.28] , [0,1] VS [0.07, 0.93] , …

Logistic Regression

$* \; 1-\sigma(s) = 1 - \frac{1}{1+e^{-s}} = \frac{e^{-s}}{1+e^{-s}}$

❑ Objective function for logistic regression

$$L = \frac{1}{N}\sum_{i=1}^{N}[\,-y_i \cdot \log\sigma(s_i) - (1-y_i)\cdot\log\big(1-\sigma(s_i)\big)\,]$$

$$= \frac{1}{N}\sum_{i=1}^{N}[\,y_i \cdot \log(1+e^{-s_i}) + (y_i-1)\cdot\log\left(\frac{e^{-s_i}}{1+e^{-s_i}}\right)]$$

$$= \frac{1}{N}\sum_{i=1}^{N}[\,y_i \cdot \log(1+e^{-s_i}) + (y_i-1)\cdot(\log e^{-s_i} - \log(1+e^{-s_i}))\,]$$

$$= \frac{1}{N}\sum_{i=1}^{N}[\,y_i \cdot \log e^{-s_i} - \log e^{-s_i} + \log(1+e^{-s_i})\,]$$

$$= \frac{1}{N}\sum_{i=1}^{N}[-y_i \cdot s_i + \log(\,e^{s_i}+1)\,] = \frac{1}{N}\sum_{i=1}^{N}[-y_i \cdot \left(\mathbf{w}^{\mathbf{T}}\mathbf{x}+b\right) + \log\left(\,e^{\mathbf{w}^{\mathbf{T}}\mathbf{x}+b}+1\right)]$$

Logistic Regression

- **Optimization**
  - $L(\mathbf{w}, b) = \frac{1}{N}\sum_{i=1}^{N}[-y_i \cdot (\mathbf{w}^{\mathbf{T}}\mathbf{x} + b) + \log\left(e^{\mathbf{w}^{\mathbf{T}}\mathbf{x}+b} + 1\right)]$
  - $\mathbf{w}^* = \underset{\mathbf{w}}{\operatorname{argmin}}\, L(\mathbf{w}, b)$, $b^* = \underset{b}{\operatorname{argmin}}\, L(\mathbf{w}, b)$
  - Solution: gradient descent!
    - Initialize $\mathbf{w}, b$
    - Repeat:
    - $w_j^{t+1} = w_j^t - \mu \frac{\partial L}{\partial w_j}(\mathbf{w}^t, b^t)$ for each $j = 1 \sim d$

      $b^{t+1} = b^t - \mu \frac{\partial L}{\partial b}(\mathbf{w}^t, b^t)$
    - $\frac{\partial L}{\partial w_j} = ?$ $\frac{\partial L}{\partial b} = ?$

Logistic Regression

$$L = \frac{1}{N}\sum_{i=1}^{N}[-y_i \cdot (\mathbf{w}^{\mathbf{T}}\mathbf{x} + b) + \log\left(e^{\mathbf{w}^{\mathbf{T}}\mathbf{x}+b}+1\right)]$$

$$\frac{\partial L}{\partial w_j} = \frac{1}{N}\sum_{i=1}^{N}[-y_i x_j + \left(\frac{x_j e^{\mathbf{w}^{\mathbf{T}}\mathbf{x}+b}}{e^{\mathbf{w}^{\mathbf{T}}\mathbf{x}+b}+1}\right)]$$

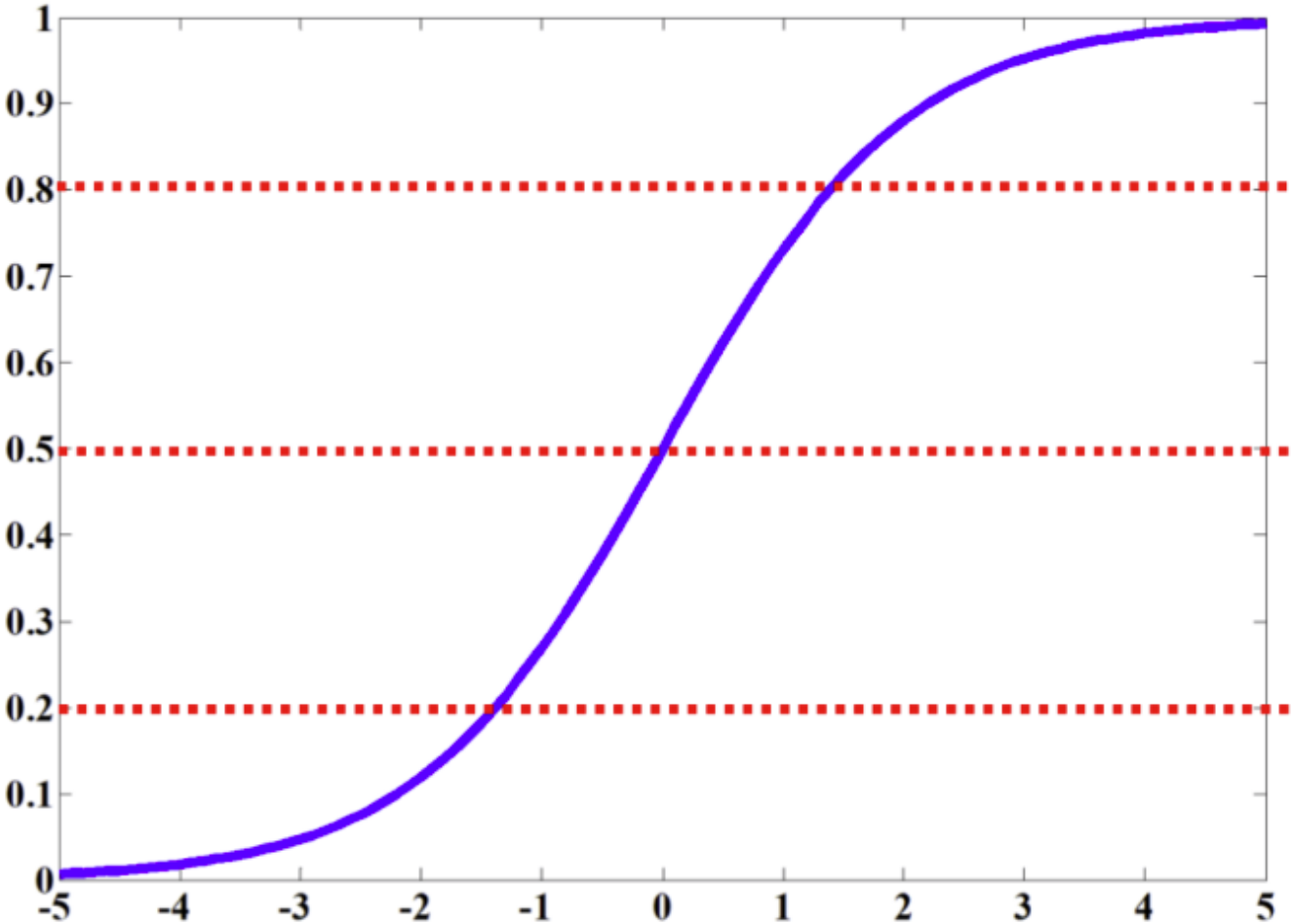
$$= \frac{1}{N}\sum_{i=1}^{N}[-y_i x_j + x_j\left(\frac{e^s}{e^s+1}\right)]$$

$$= \frac{1}{N}\sum_{i=1}^{N}[-y_i x_j + x_j\sigma(s_i)] = \frac{1}{N}\sum_{i=1}^{N}[\, x_j \cdot (\sigma(s_i) - y_i)\,]$$

$$\frac{\partial L}{\partial b} = \frac{1}{N}\sum_{i=1}^{N}[-y_i + \left(\frac{e^{\mathbf{w}^{\mathbf{T}}\mathbf{x}+b}}{e^{\mathbf{w}^{\mathbf{T}}\mathbf{x}+b}+1}\right)] = \frac{1}{N}\sum_{i=1}^{N}[\,\sigma(s_i) - y_i\,]$$

Logistic Regression Summary

- **Inference (after training)**
  - Classify each test case based on your own threshold



When data is biased towards positive cases

Typically used: 0.5

When data is biased towards negative cases

Logistic Regression Summary

- **Regressing probability**
  - Put our signal $s = \mathbf{w}^{\mathbf{T}}\mathbf{x} + b$ into $\frac{1}{1+e^{-s}}$ bounded within 0 and 1
- **Cross entropy loss**

$$L = -y \cdot \log\sigma(s) - (1-y) \cdot \log\big(1 - \sigma(s)\big)$$

- **Loss function**

$$L(\mathbf{w}, b) = \frac{1}{N}\sum_{i=1}^{N}[-y_i \cdot \left(\mathbf{w}^{\mathbf{T}}\mathbf{x} + b\right) + \log\left(\, e^{\mathbf{w}^{\mathbf{T}}\mathbf{x}+b} + 1\right)]$$

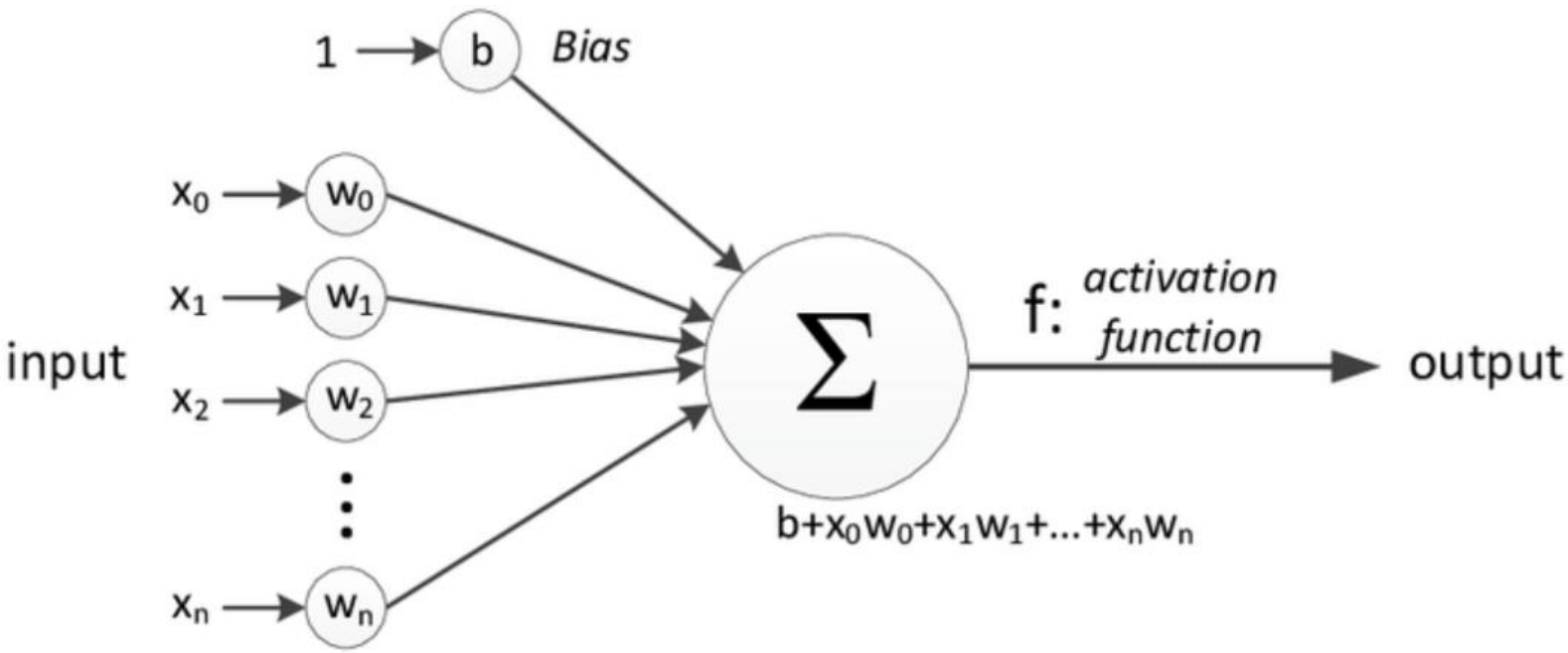
- **Gradient**

$$\frac{\partial L}{\partial w_j} = \frac{1}{N}\sum_{i=1}^{N}[\, x_j \cdot (\sigma(s_i) - y_i)\,], \text{ for } i = 1 \sim d$$

$$\frac{\partial L}{\partial b} = \frac{1}{N}\sum_{i=1}^{N}[\, \sigma(s_i) - y_i\,]$$

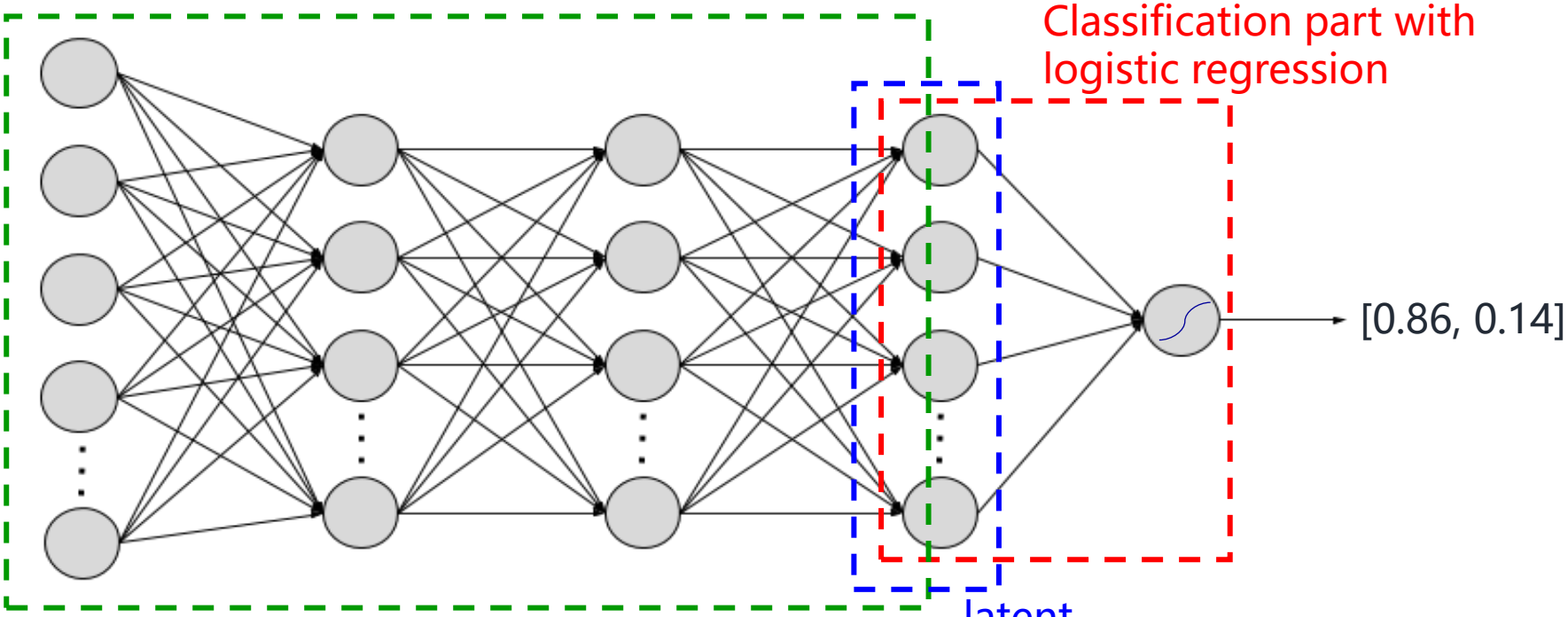
Future Topics

- **A "Perceptron" model**
  - A perceptron model (or, an artificial neuron) with **sigmoid activation** and **cross-entropy loss function** is identical to the logistic regression



Future Topics

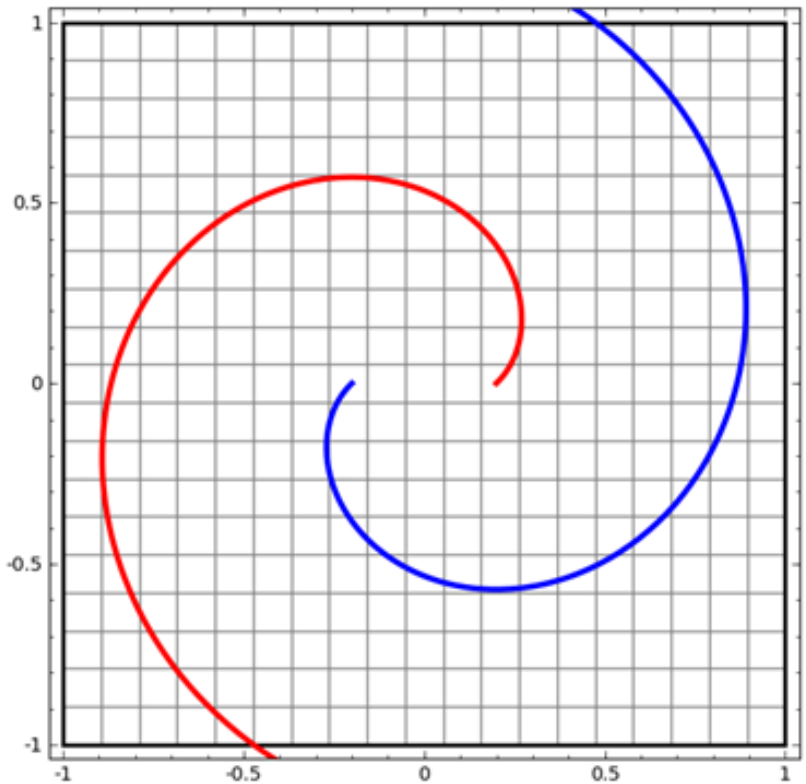
- **Multi-layer neural network**



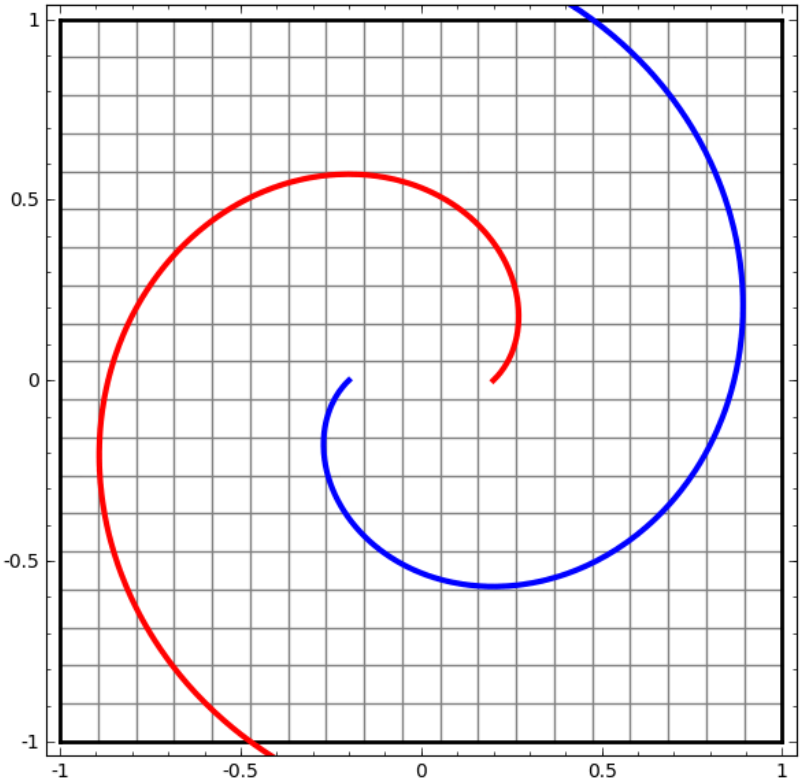
Data transformation
Data projection to latent space
Feature learning
Embedding

Future Topics

- **Multi-layer neural network**



Input data

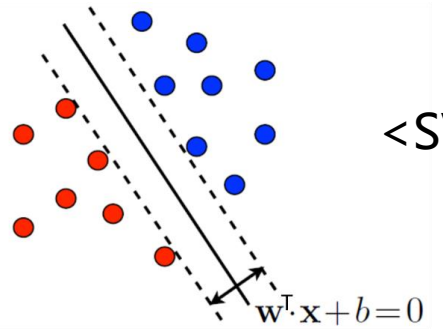


Linear and non-linear transformation

Future Topics

- **SVM**
  - Accepting the same signal, processing it in a different way

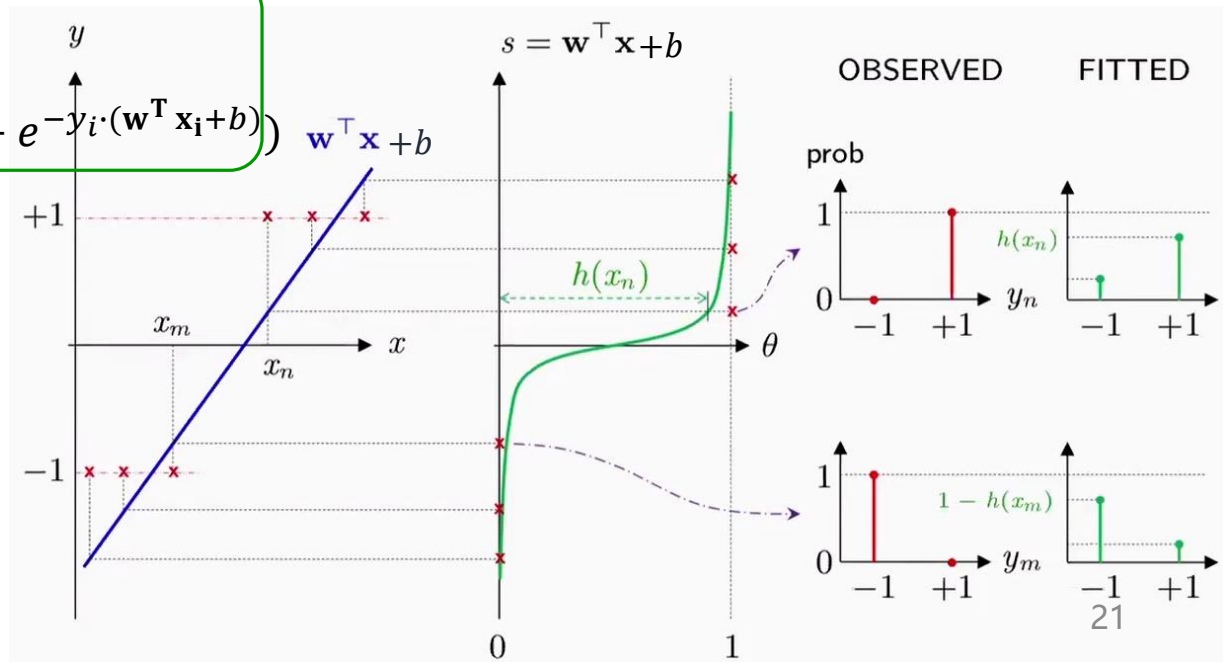
$$\min \ \frac{1}{2}\|\mathbf{w}\|^2$$
$$s.t. \ \ y_i(\mathbf{w}^{\mathrm{T}}\mathbf{x}_i + b) \geq 1, \ \ \forall i \in [\mathbf{1}, \boldsymbol{N}]$$



<SVM>

$$\mathbf{w}^* = \underset{\mathbf{w}}{\operatorname{argmin}}\ L(\mathbf{w})$$

where $L(\mathbf{w}) = \Sigma\log(1 + e^{-y_i\cdot(\mathbf{w}^{\mathbf{T}}\mathbf{x_i}+b)})$



<logistic regression>

Thank You